June 30, 2023

Reflection 1 : Due tonight by midnight

Conceptual HW 1 : Due Sunday by midnight

No Class Monday!

Monday office hours moved to Wed

No Lab Tuesday!

Written HW 1 : Due Thursday by midnight

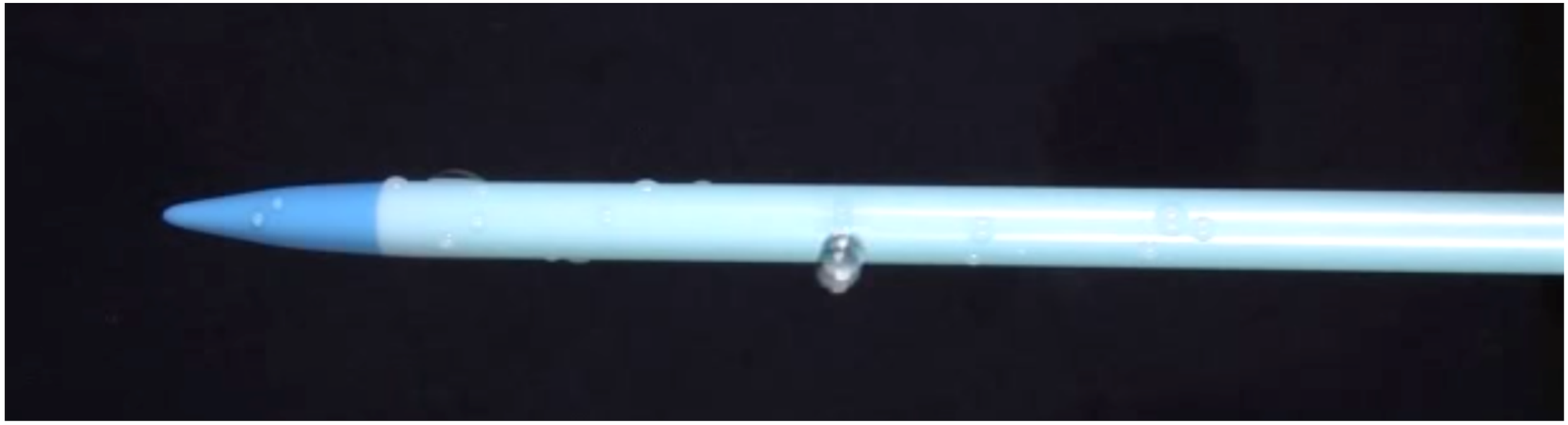
Office Hours

Mondays Noon-2pm
Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	June 26 Intro, V2 CH5	June 27 Lab W1 - Elec Force	June 28 V2 CH5	June 29 V2 CH5 Studio : For Field	June 30 V2 CH5 Reflection
Week 2	July 3 No Physics 213 Class	July 4 No Classes! OSU Closed.	July 5 V2 CH 5	July 6 V2 CH 6 Homework 1 Studio : Field Calcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Using E	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Gauss's Law	July 14 V2 CH 7 Reflection

Chapter 5

Electric Charges and Fields



Science off the Sphere: Knitting Needle Experiment

NASA: Johnson Space Center (Houston)

https://www.youtube.com/watch?v=qHrBhgwq__Q

Continuous Charge Distributions

§ 5.5
Lecture 04

Steps for Continuous Fields

Draw a picture and setup coordinates

Chose a tiny charge dq on the object (no special locations)

Figure out the <u>distance vector</u> $\vec{r}$ from dq to your point

Use the distance vector to get distance and direction

Write down dE for that point : $d\vec{E} = K \frac{dq}{r^2} \hat{r}$

Change dq to a distribution and differentials

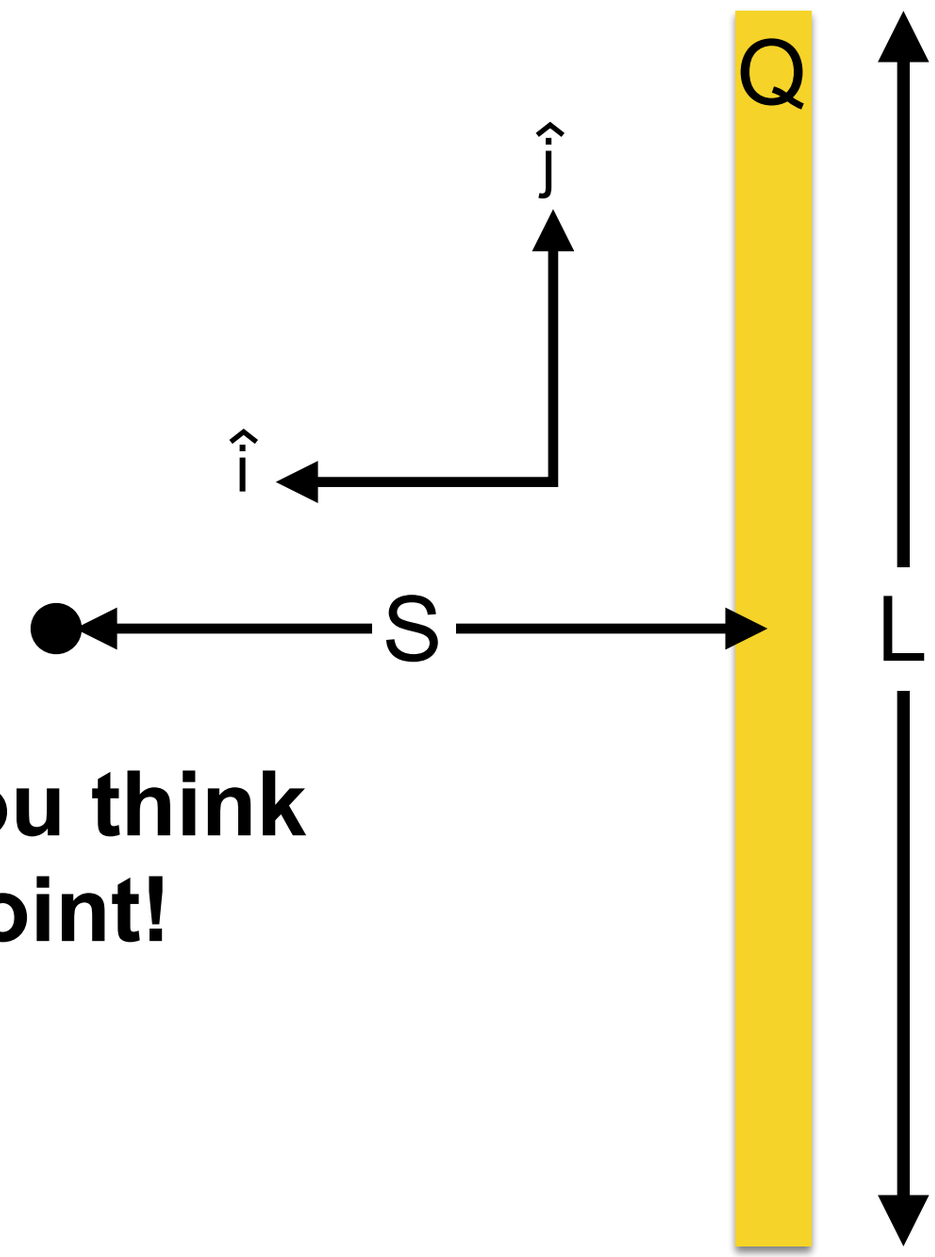
Figure out the limits

Integrate and simplify?

Line of Charge

Draw a Picture & Setup Coordinates

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?



Quickly sketch what you think E will look like at the point!

Line of Charge

Use the distance vector to get distance and direction

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

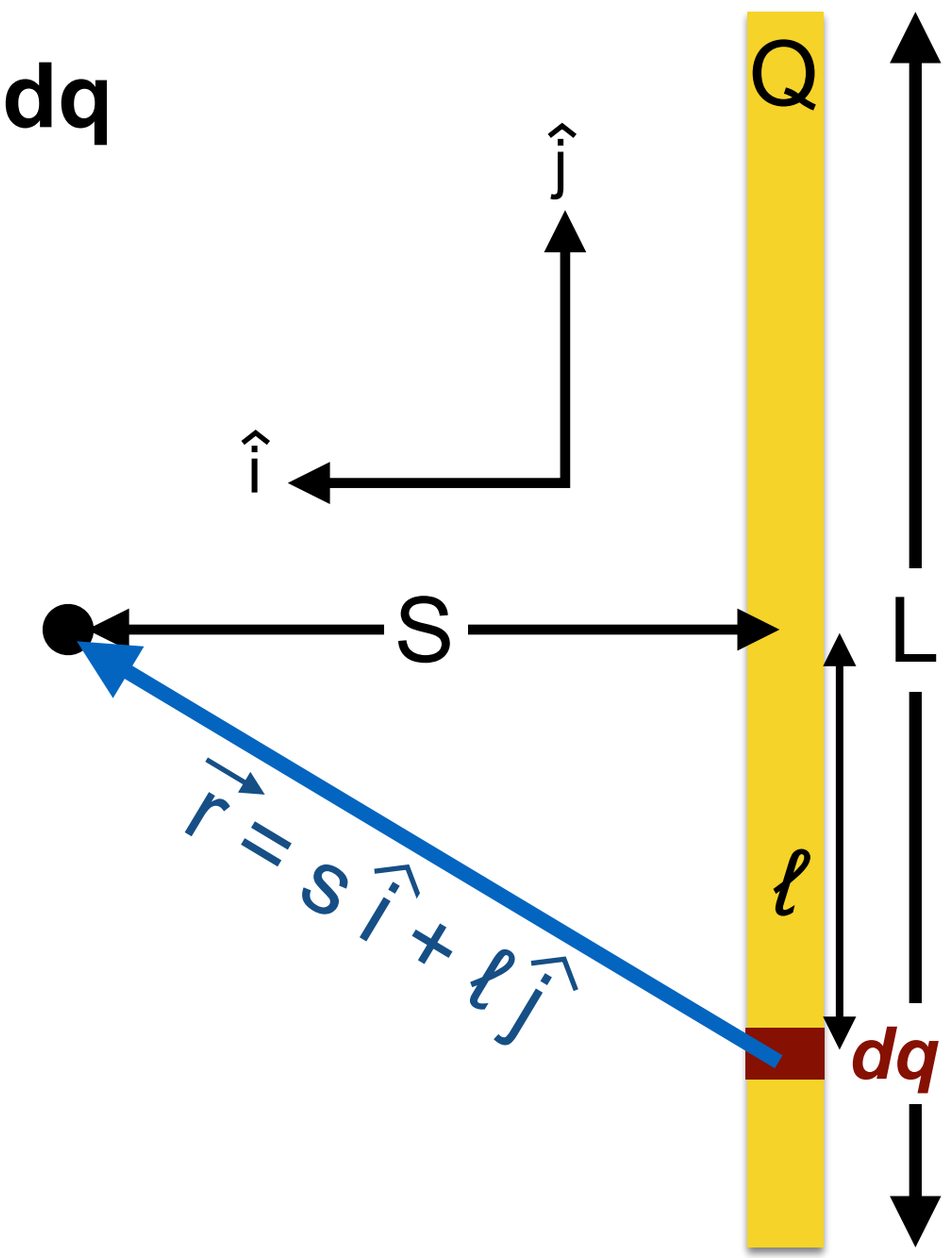
Determine the unit vector from dq to the point.

Distance: $r = \sqrt{(r_x)^2 + (r_y)^2}$

$r = \sqrt{s^2 + \ell^2}$

Direction:

$$\hat{r} = \frac{\vec{r}}{r} = \frac{s\,\hat{i} + \ell\,\hat{j}}{\sqrt{s^2 + \ell^2}}$$



Line of Charge

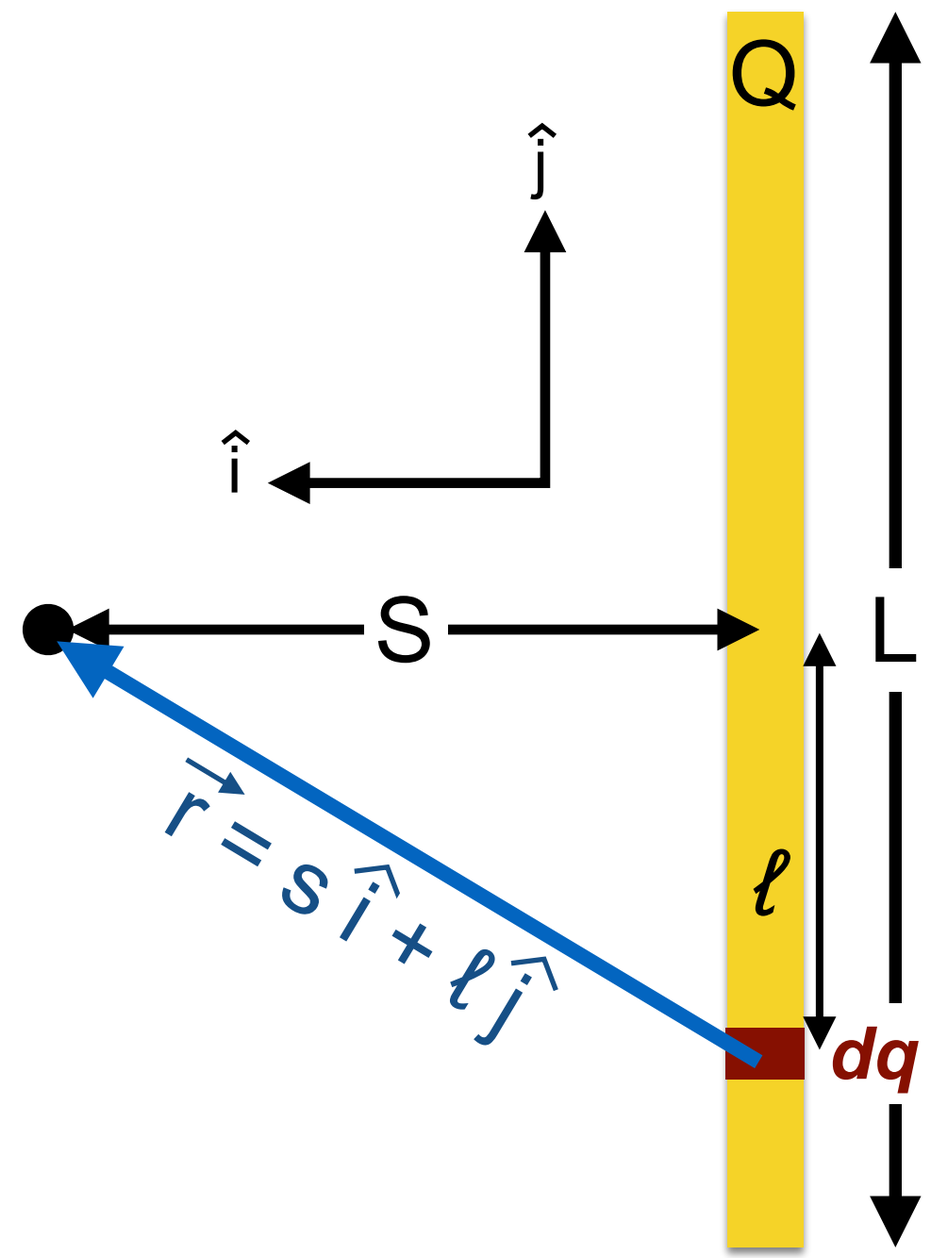
Write down dE for that point

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$d\vec{E} = K \frac{dq}{r^2} \hat{r}$$

$$d\vec{E} = K \frac{dq \quad (s\hat{i} + \ell\hat{j})}{(s^2+\ell^2)^{3/2}}$$

This is the electric field created by tiny charge dq on the line



Line of Charge

Change dq to a distribution and differentials

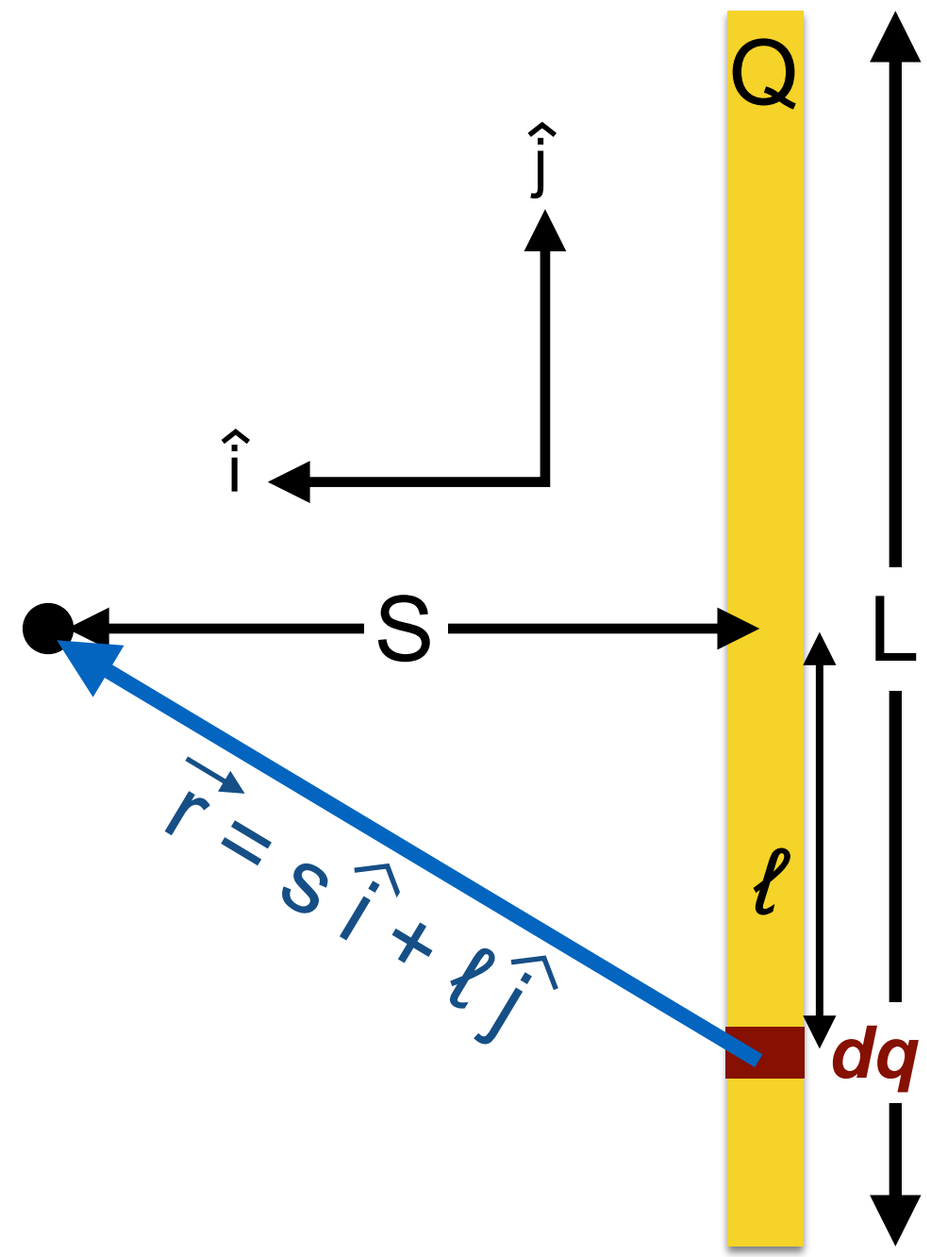
What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$d\vec{E} = K \frac{dq\,(s\,\hat{i} + \ell\,\hat{j})}{(s^2+\ell^2)^{3/2}}$$

A line of charge : $dq = \lambda\, d\ell$

$$d\vec{E} = K \frac{\lambda\, d\ell\,(s\,\hat{i} + \ell\,\hat{j})}{(s^2+\ell^2)^{3/2}}$$

This is the electric field created by charged line segment $d\ell$



Line of Charge

Figure out the limits

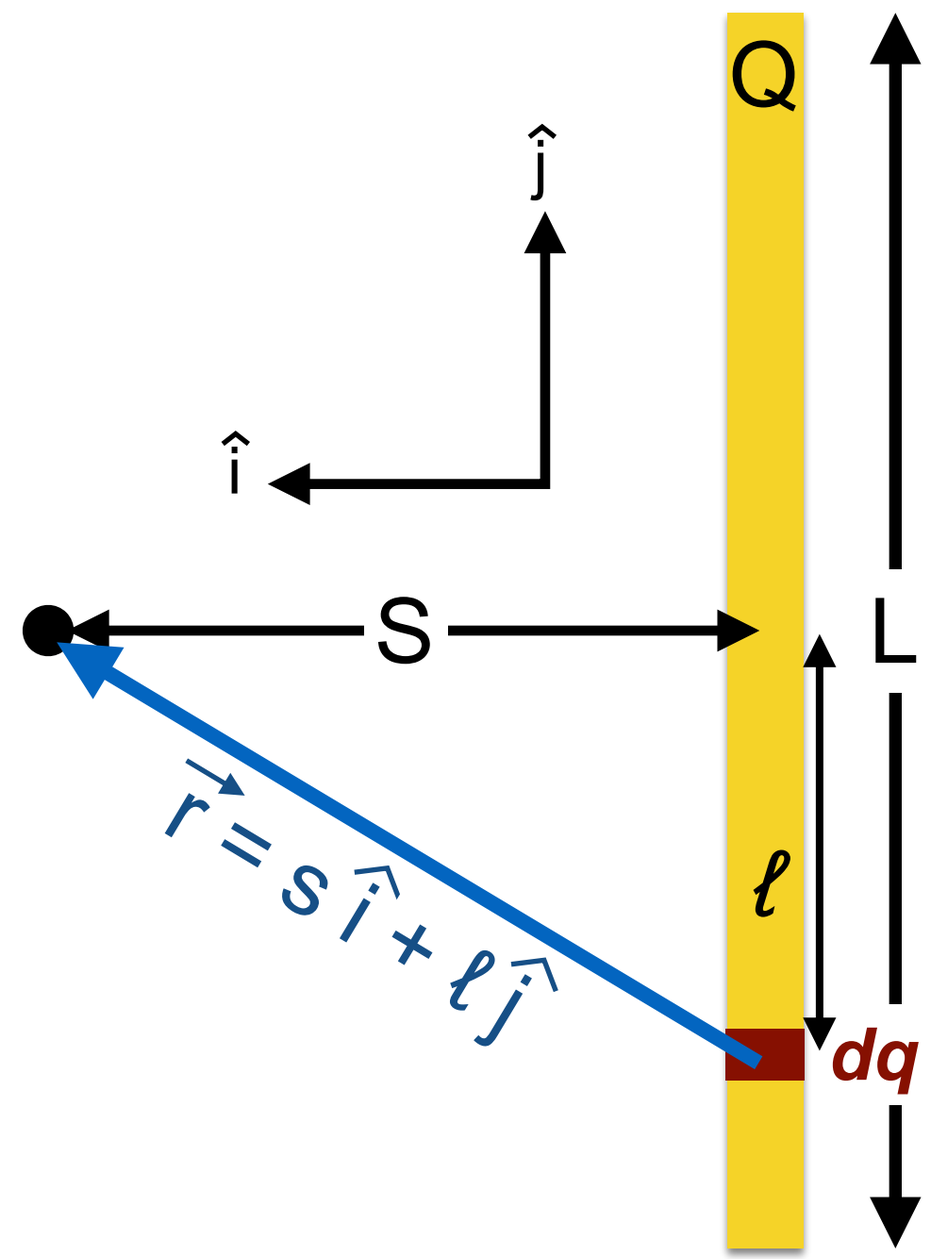
What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$d\vec{E} = K\,\frac{\lambda\, d\ell\,(s\,\hat{i} + \ell\,\hat{j})}{(s^2+\ell^2)^{3/2}}$$

$$\vec{E} = \int d\vec{E}$$

$$\vec{E} = \int K\,\frac{\lambda\, d\ell\,(s\,\hat{i} + \ell\,\hat{j})}{(s^2+\ell^2)^{3/2}}$$

We have to integrate along the whole length of the line!

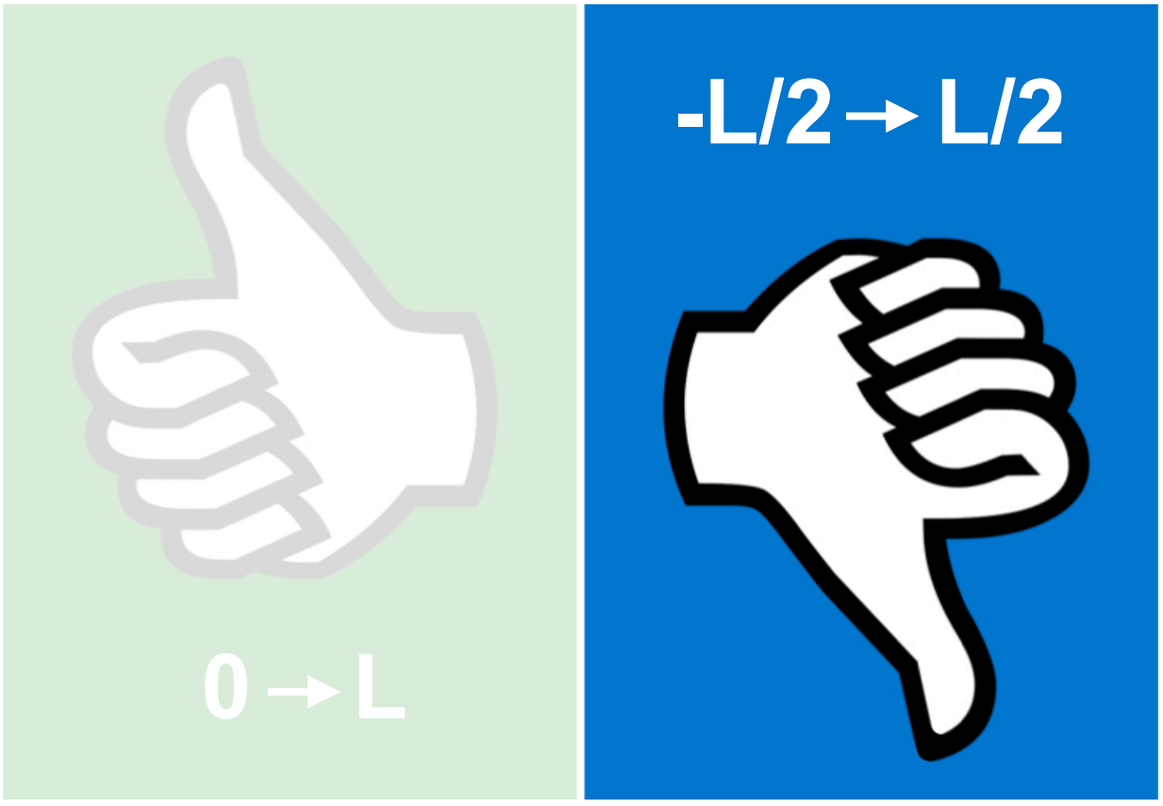


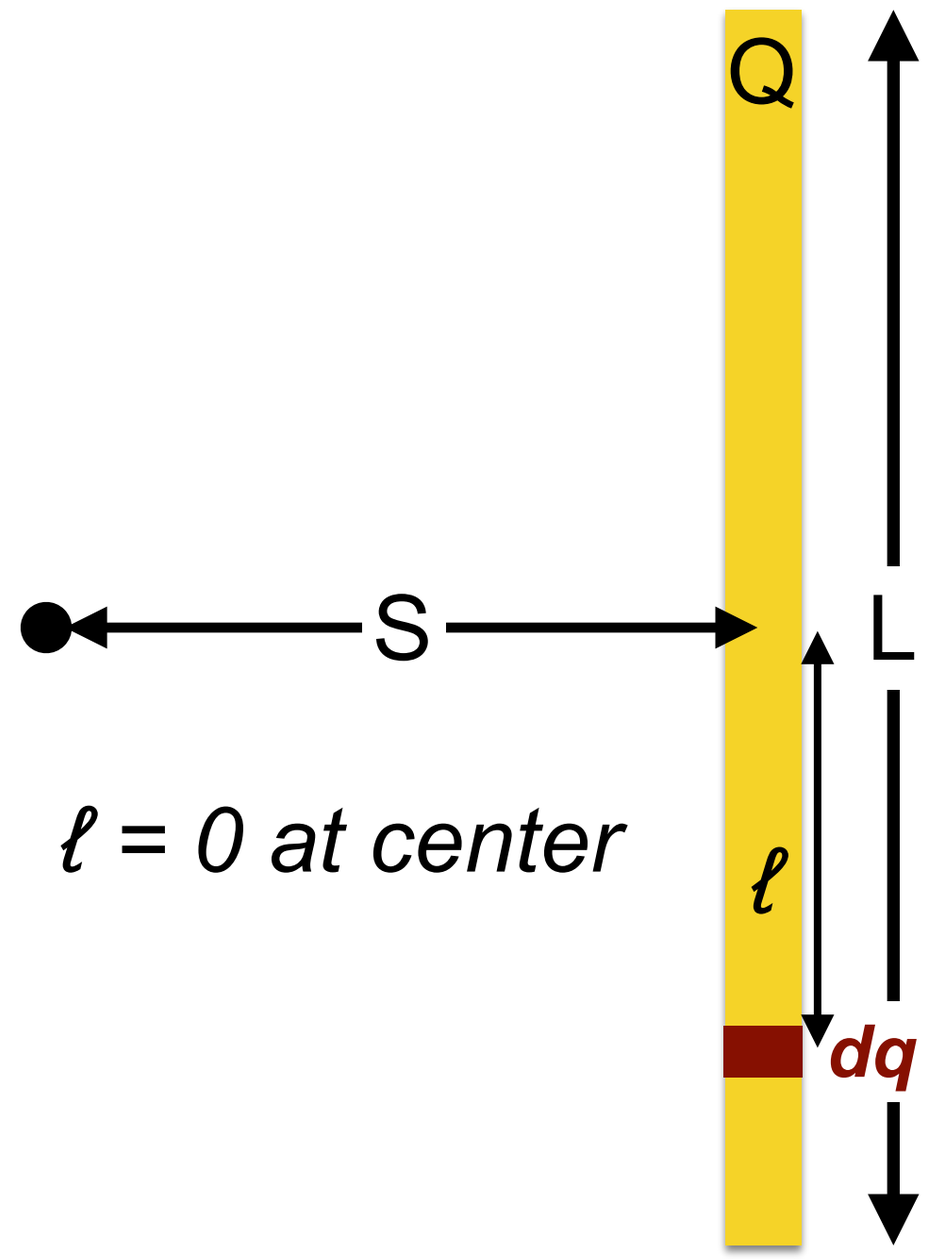
Line of Charge

Limits of integral

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

To integrate along this **whole line**, we should use the limits…





0. Don't wait on my response

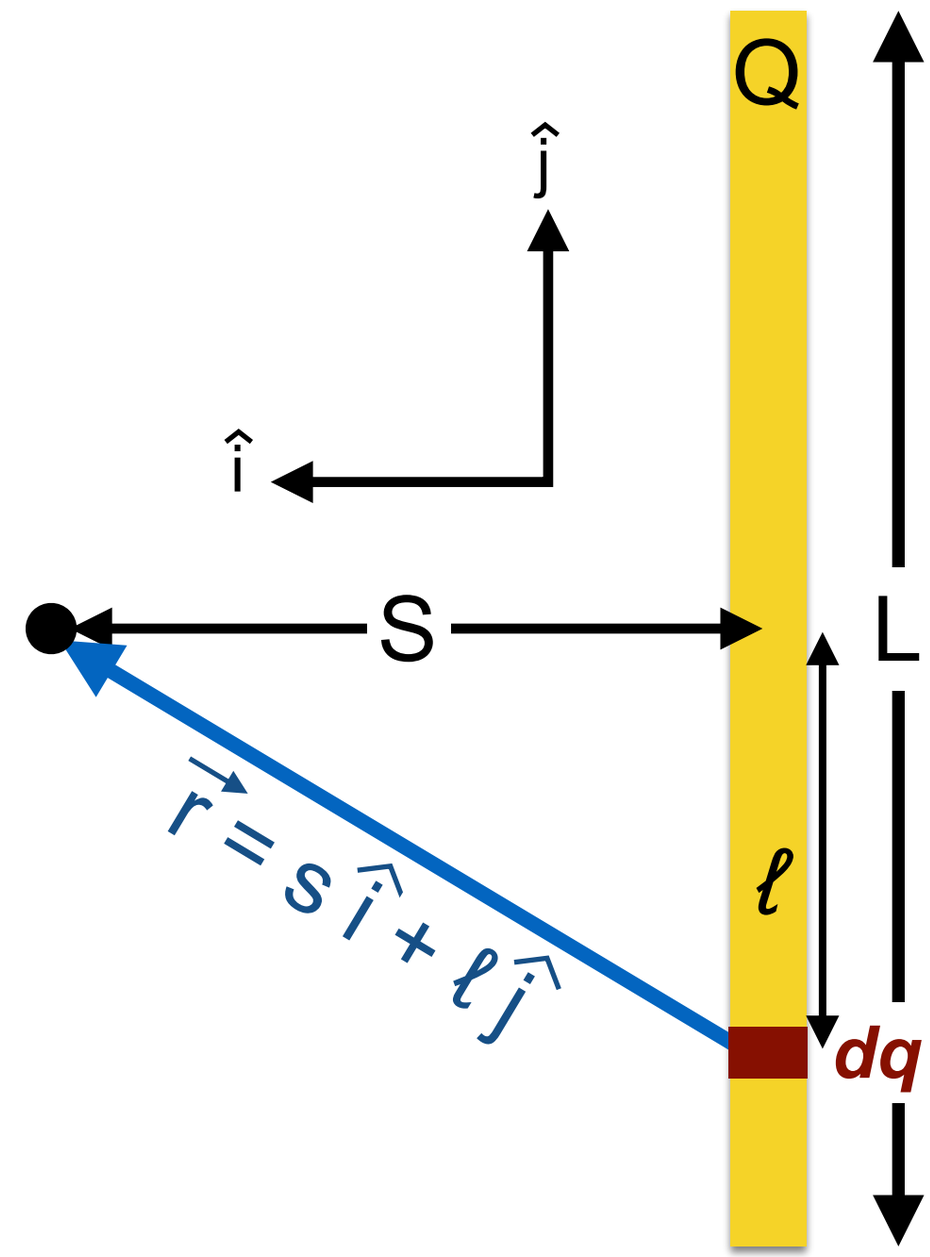
Line of Charge

Integrate?

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$\vec{E} = \int_{-L/2}^{L/2} K \frac{\lambda \, d\ell \;\; (s\,\hat{i} + \ell\,\hat{j})}{(s^2+\ell^2)^{3/2}}$$

To integrate a vector, integrate each component separately.



Line of Charge

X-Integral Lookup Table

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$E_x = K \lambda s \int_{-L/2}^{L/2} \frac{d\ell}{(s^2+\ell^2)^{3/2}}$$

$$\int \frac{dx}{(x^2 \pm a^2)^{3/2}} = \frac{\pm x}{a^2\sqrt{x^2 \pm a^2}}$$

$x \rightarrow \ell$

$a \rightarrow s$

$$E_x = K \lambda s \left. \frac{+\ell}{s^2\sqrt{\ell^2+s^2}} \right|_{-L/2}^{L/2}$$

Solve me!

$$E_x = \frac{K \lambda L}{s\sqrt{L^2/4+s^2}}$$

Line of Charge

X-Integral Lookup Table

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$E_y = K\lambda \int_{-L/2}^{L/2} \frac{\ell \, d\ell}{(s^2+\ell^2)^{3/2}}$$

$$\int \frac{x\,dx}{(x^2 \pm a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 \pm a^2}}$$

$x \rightarrow \ell$

$a \rightarrow s$

$$E_y = K\lambda \left. \frac{-1}{\sqrt{\ell^2+s^2}} \right|_{-L/2}^{L/2}$$

Solve me!

$$E_y = 0$$

Also note : dE_y was an odd function with symmetric limits!

Line of Charge

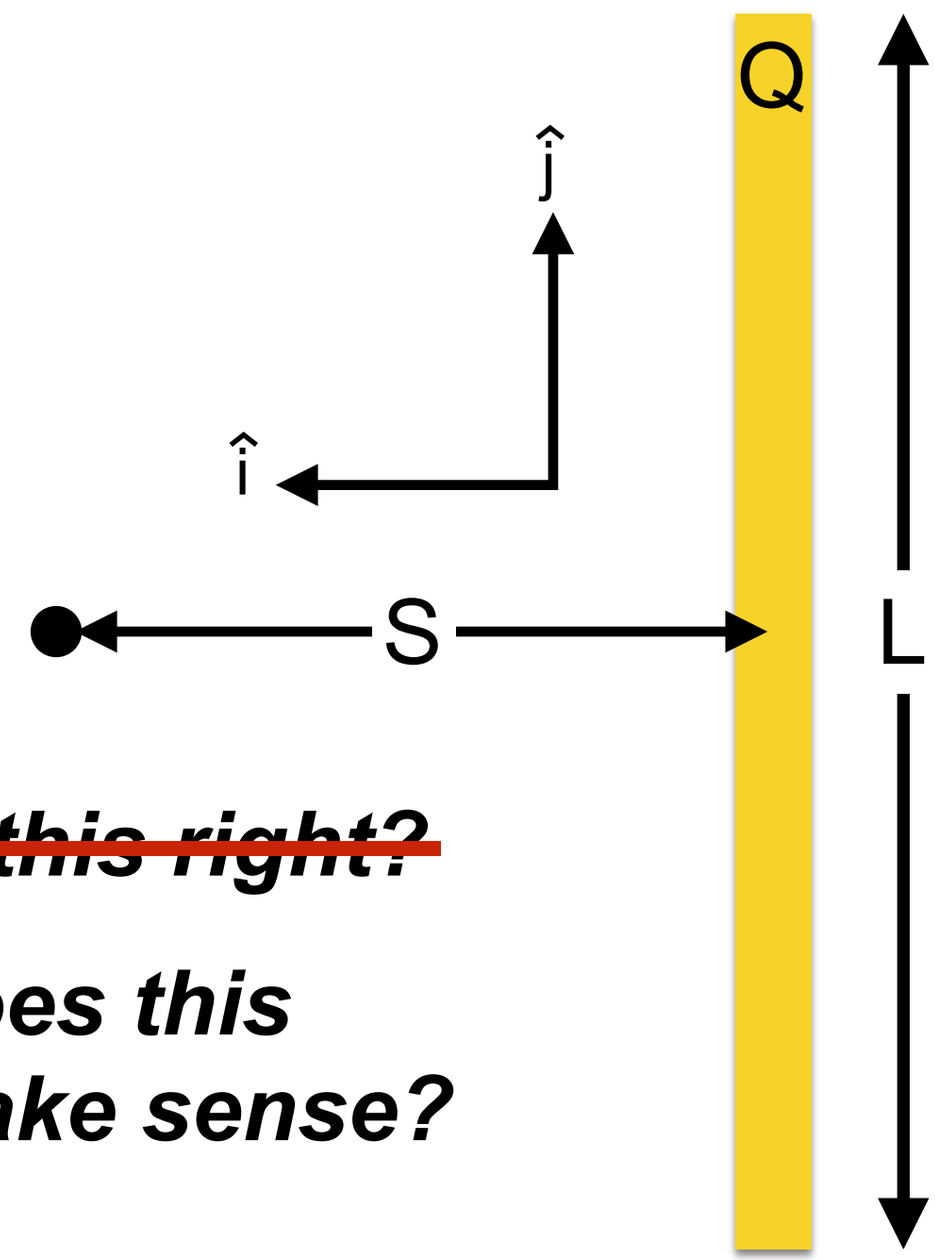
Answer

What is the electric field at distance s from the center of a line of charge with total length L and total charge Q?

$$\vec{E} = \frac{K \lambda L \hat{i}}{s\sqrt{L^2/4+s^2}}$$

$$\vec{E} = \frac{K (Q/L) L \hat{i}}{s\sqrt{L^2/4+s^2}}$$

$$\vec{E} = \frac{K Q \hat{i}}{s\sqrt{L^2/4+s^2}}$$



~~***Is this right?***~~

Does this make sense?

Line of Charge

Check Units!

$$\vec{E} = \frac{K\,Q\,\hat{i}}{s\sqrt{L^2/4+s^2}}$$

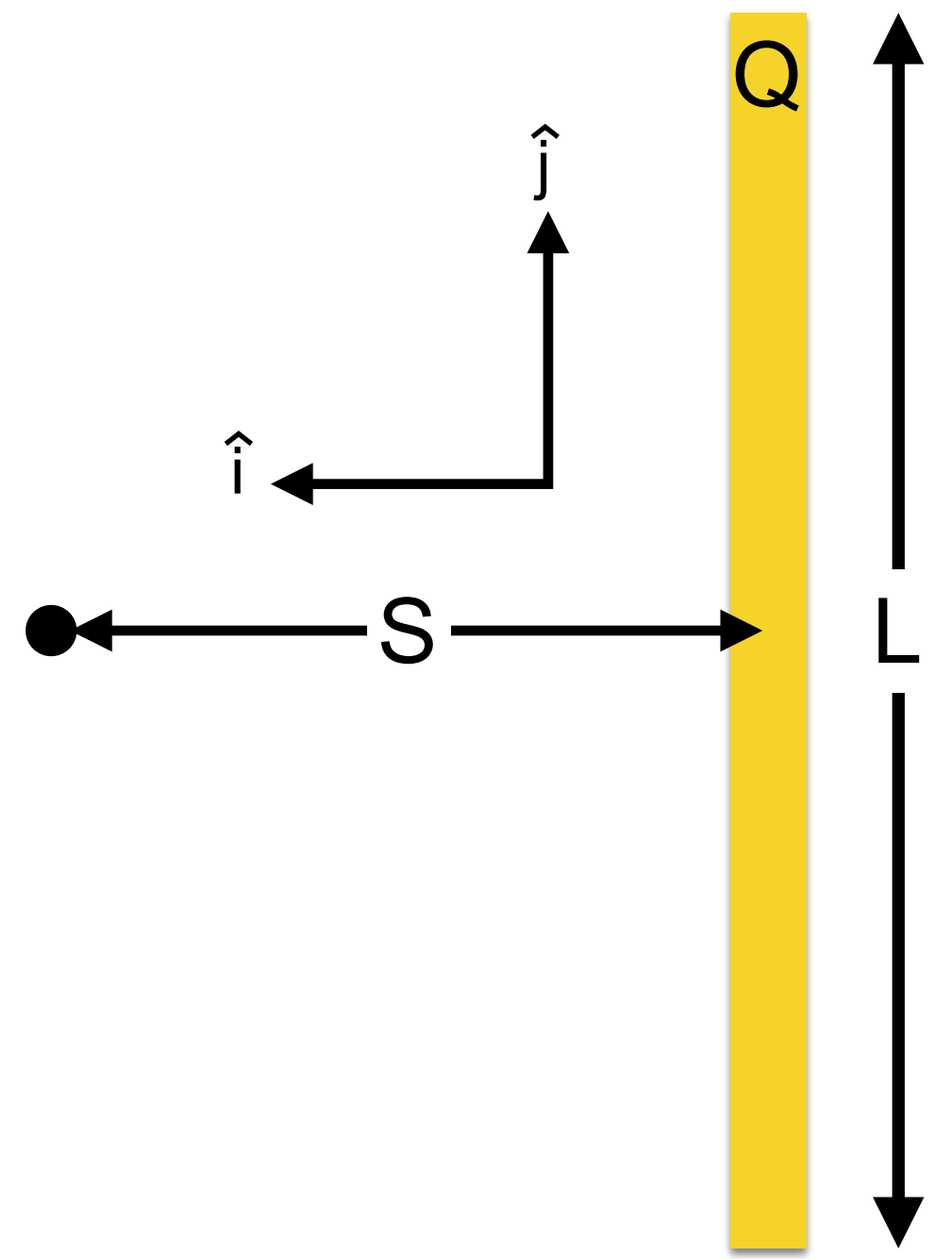
$K \sim N\,m^2/C^2$

$Q \sim C$

$s \sim m$

$L \sim m$

$$E \sim \frac{(N\cancel{m^2}/\cancel{C^2})\,(\cancel{C})}{(\cancel{m})\cancel{\sqrt{m^2/4+m^2}}} \sim N/C$$



Line of Charge

Check Direction!

$$\vec{E} = \frac{K Q \hat{i}}{s\sqrt{L^2/4+s^2}}$$

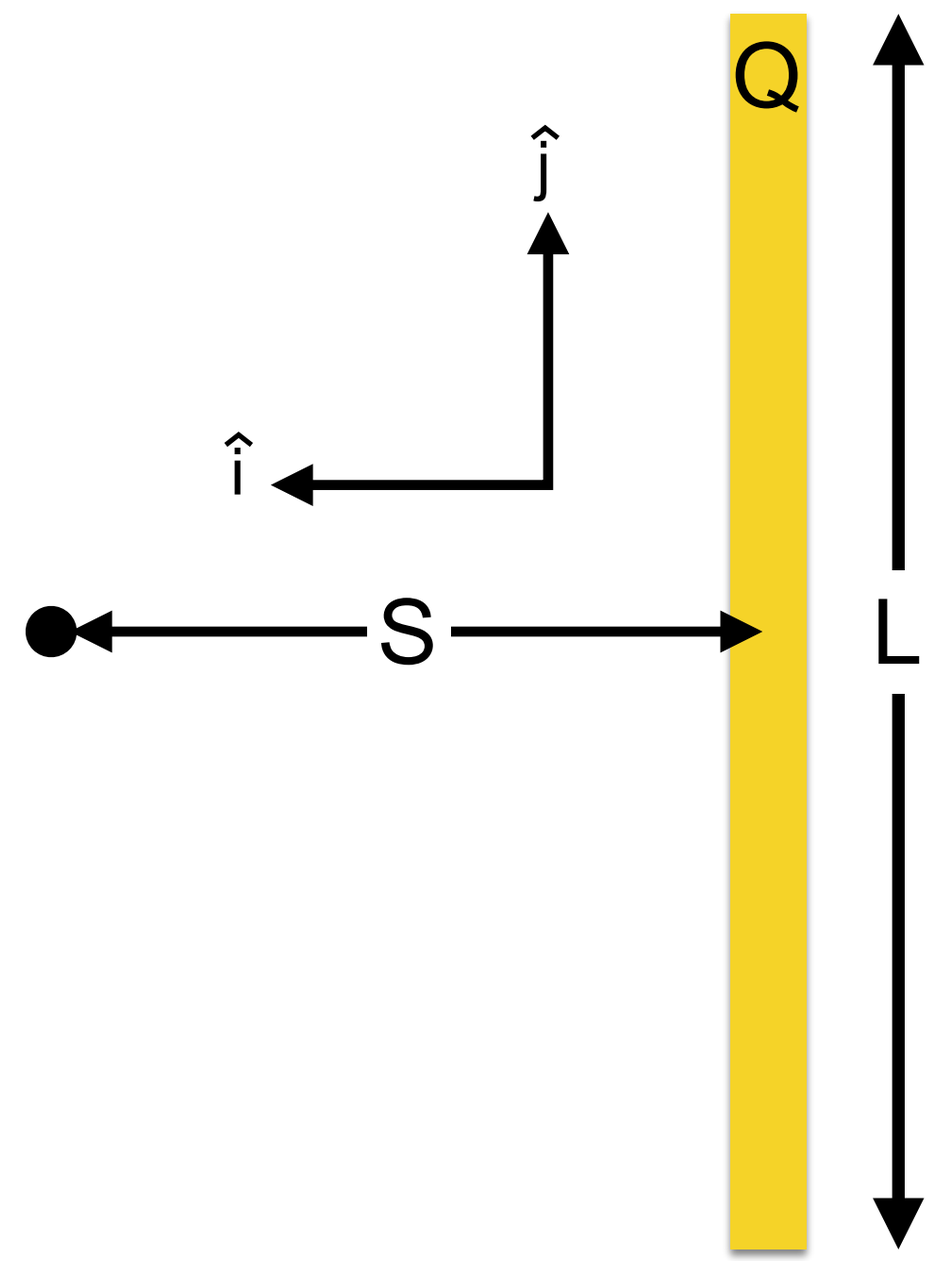
If Q>0, what direction is E according to our equation?

If Q<0, what direction is E according to our equation?

1. Up
2. Down
3. Right
4. Left
5. More than one of these

Q<0

Q>0

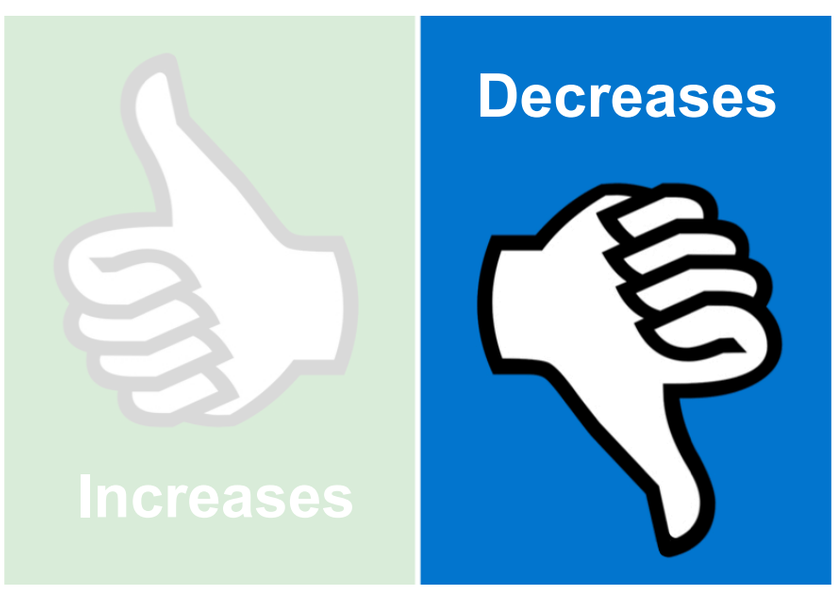


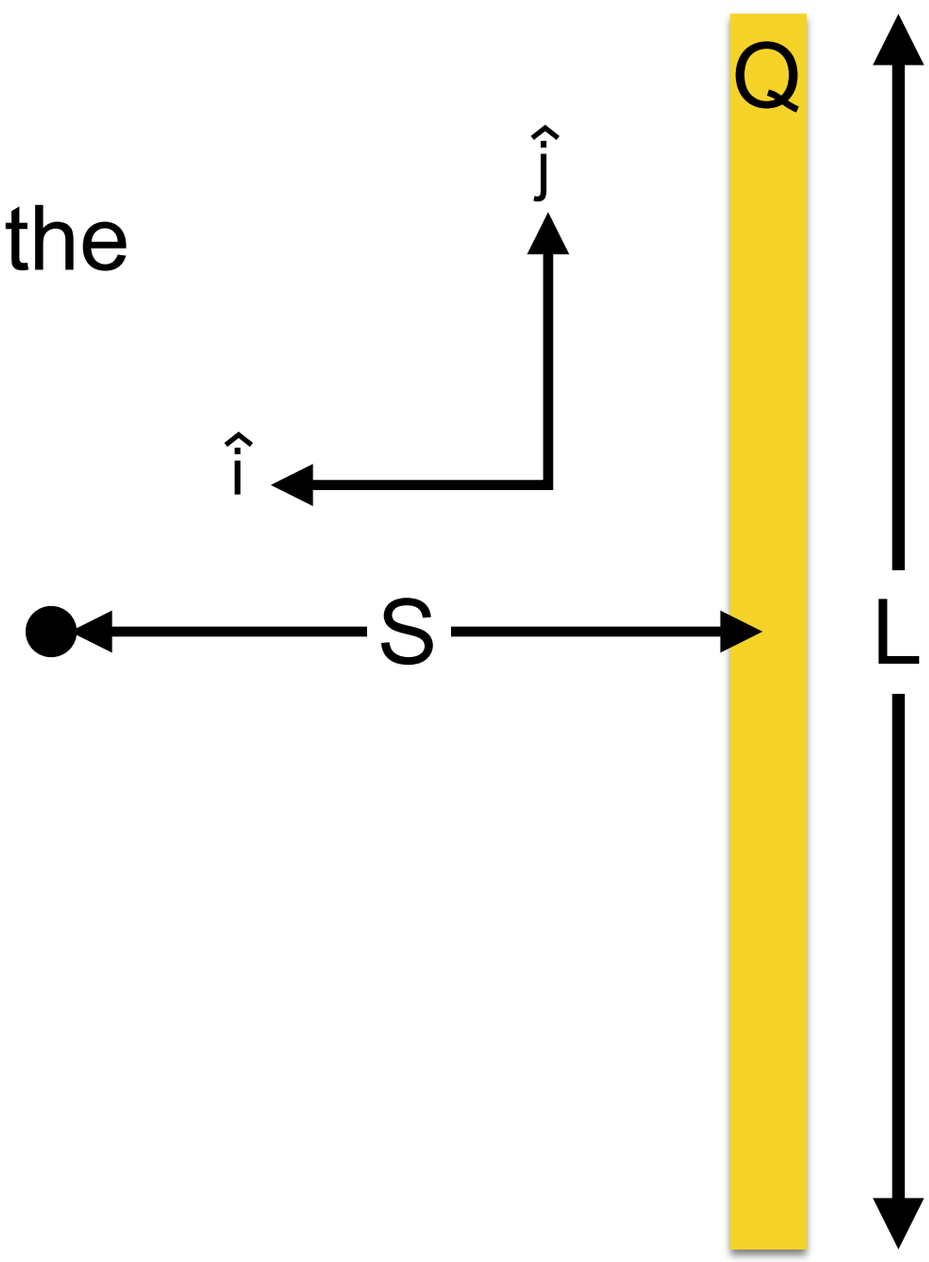
Line of Charge

Check Behavior!

$$\vec{E} = \frac{K Q \hat{i}}{s\sqrt{L^2/4 + s^2}}$$

Farther from the line, how does the electric field strength change?



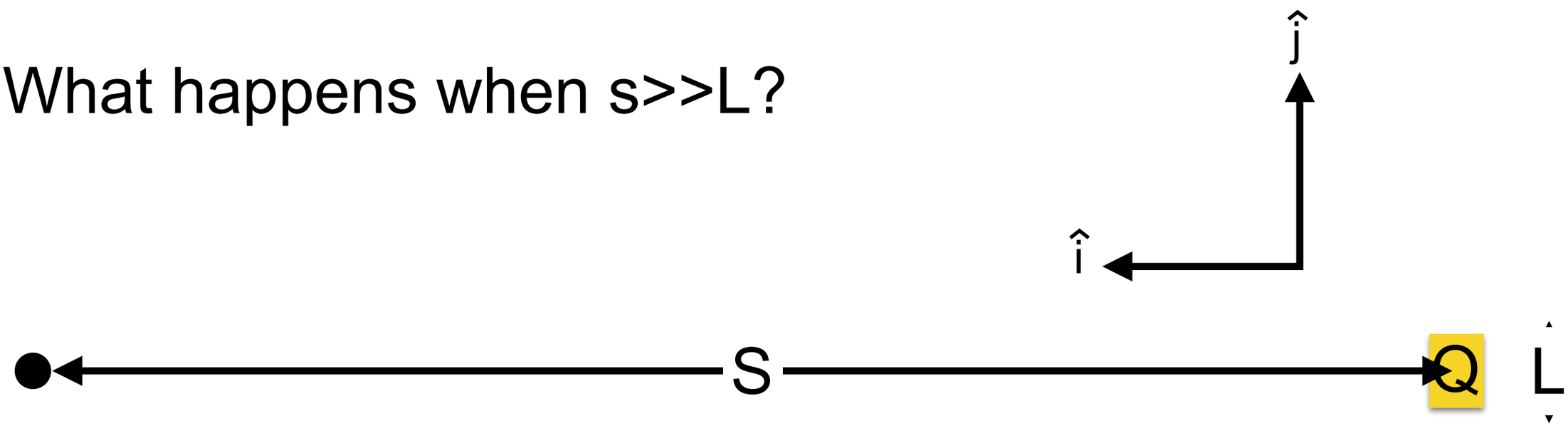


Line of Charge

Check Limiting Behavior!

$$\vec{E} = \frac{K\,Q\,\hat{i}}{s\sqrt{L^2/4+s^2}}$$

What happens when s>>L?



$$\sqrt{L^2/4+s^2} \longrightarrow \sqrt{0^2+s^2} = s$$

$$\vec{E} = \frac{K\,Q\,\hat{i}}{s\sqrt{L^2/4+s^2}} \longrightarrow \frac{K\,Q\,\hat{i}}{s\,s} = \frac{K\,Q\,\hat{i}}{s^2}$$

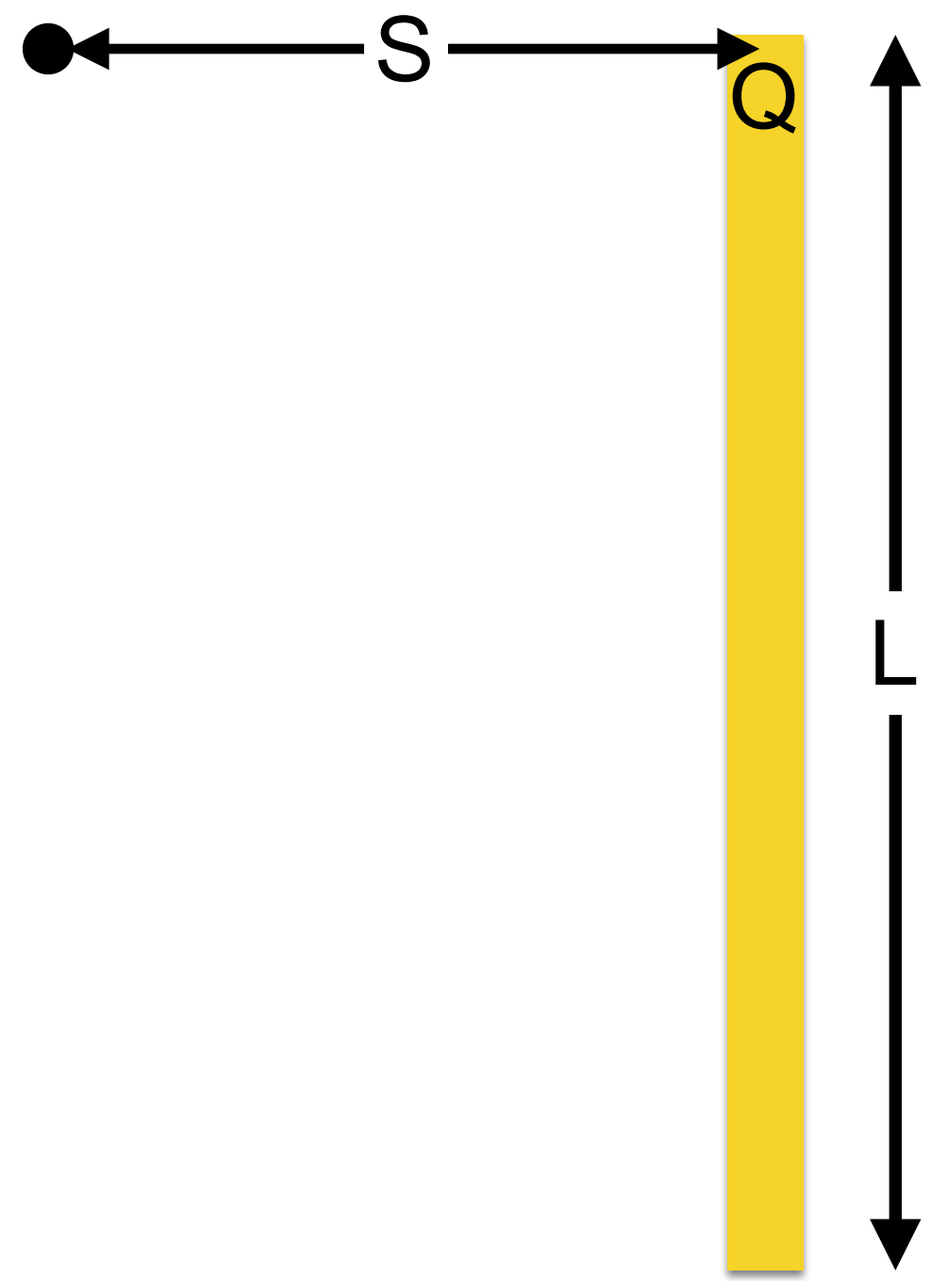
Looks like a single charge!

Line of Charge 2

Perpendicular from from End

Consider this problem:

What is the electric field at distance s perpendicular from the end of a line of charge with total length L and total charge +Q?



Spend a few minutes thinking, and then write about how you think the calculation will be *different* from the calculation we did in class.

https://forms.gle/FzxEk3pHK6gkJhQC9

Pre-class Slides

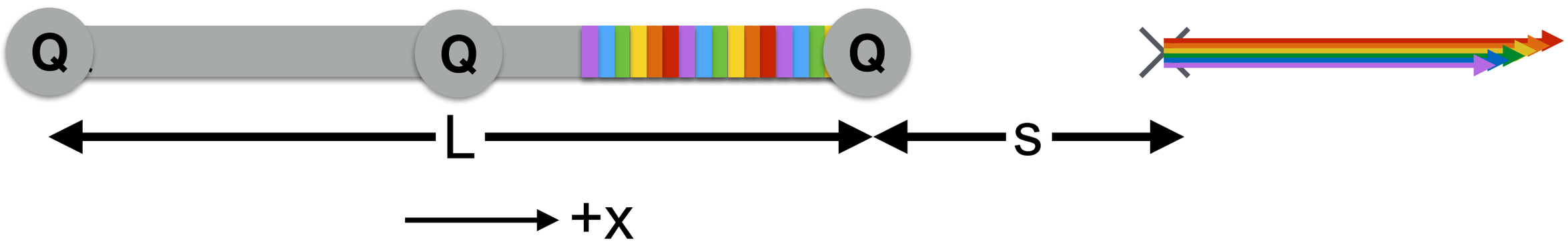
Continuous Charge Distributions

§ 5.5
Lecture 04

Youtube Playlist

Line of Charge

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$\cancel{\vec{E} = k\,Q / s^2\,\hat{i}}$

$\cancel{\vec{E} = k\,Q / (s+L)^2\,\hat{i}}$

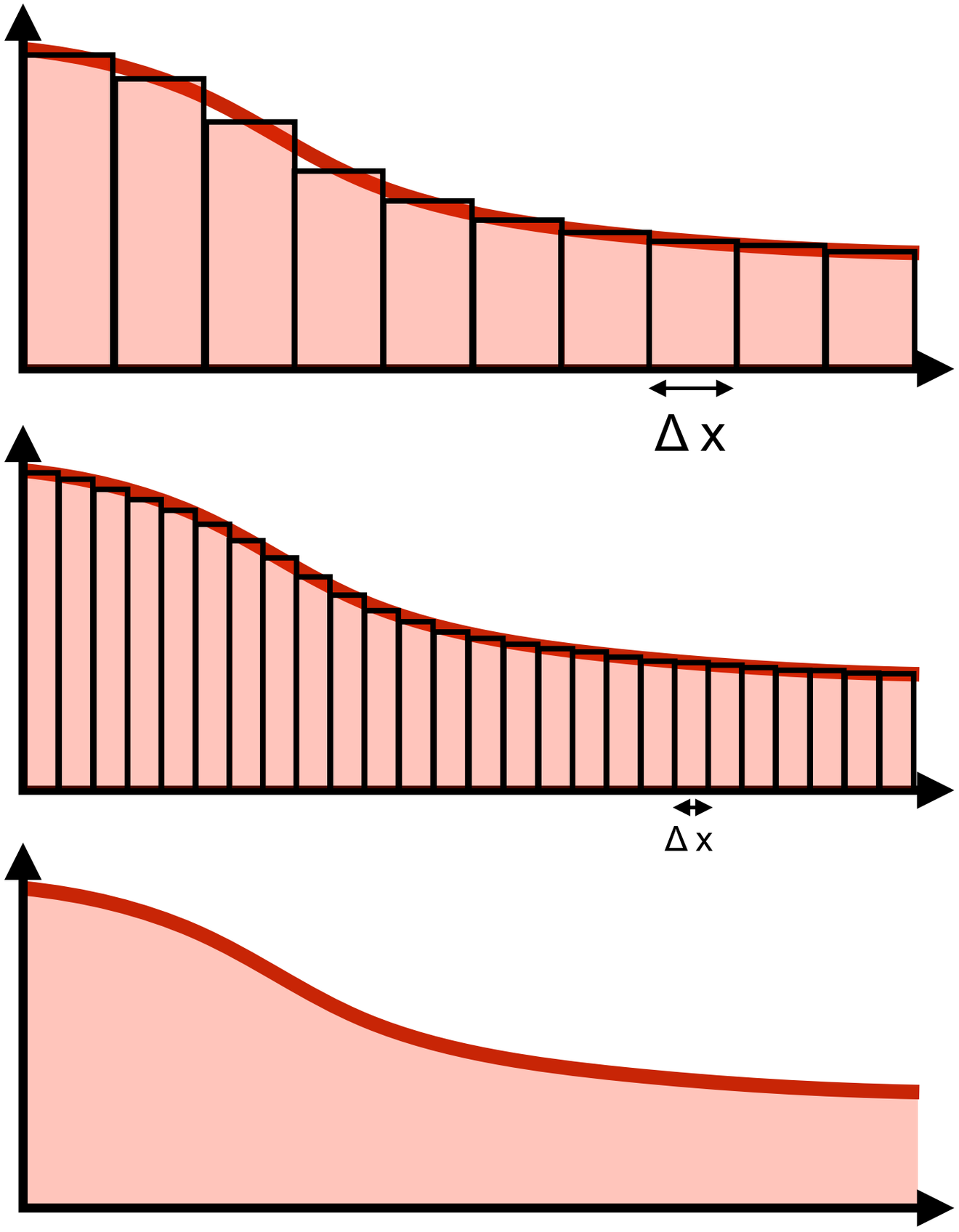
$\cancel{\vec{E} = k\,Q / (s+L/2)^2\,\hat{i}}$

Average of r^2 is not $r/2$

$$\vec{E} = K\,\frac{Q}{r^2}\,\hat{r}$$

$$\vec{E} = \sum_i \vec{E}_i$$

Review : Area Under The Curve



Reimann Sum

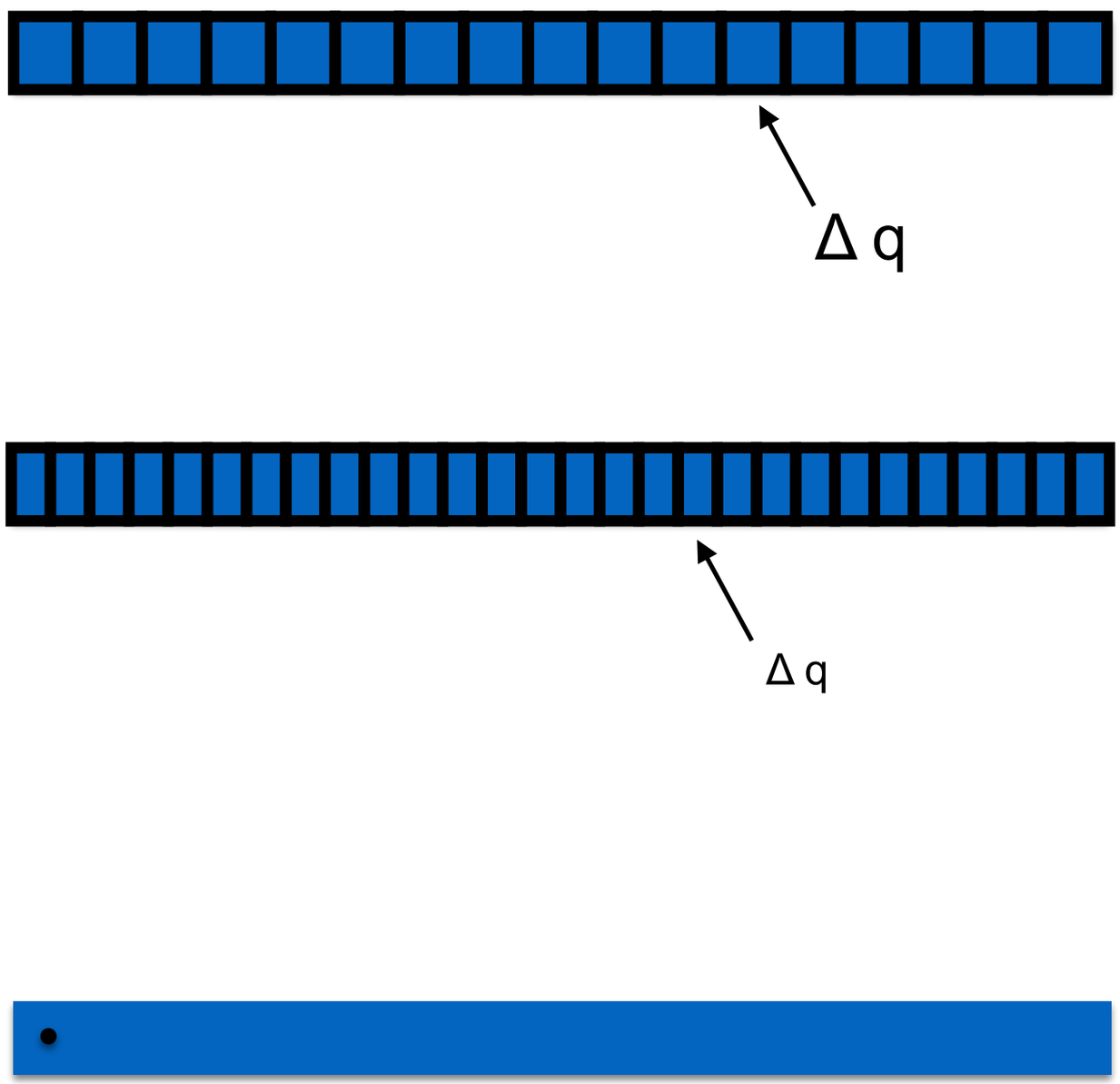
↓

Infinitely small width, dx

↓

Integral

Plan : Electric Field of an Object



Vector Sum of ΔE

↓

Infinitely small charge, dq

↓

Integral over dE

Charge Distributions

Line of Charge

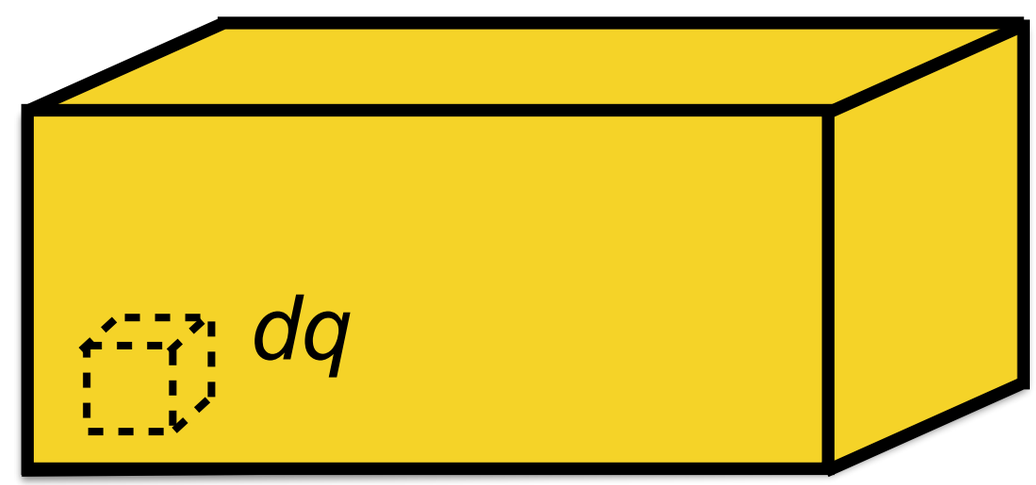
$\lambda_{uniform} = Q / L$

$dq = \lambda \, d\ell$

Surface Charge

$\eta_{uniform} = Q / A \quad dq = \eta \, dA$

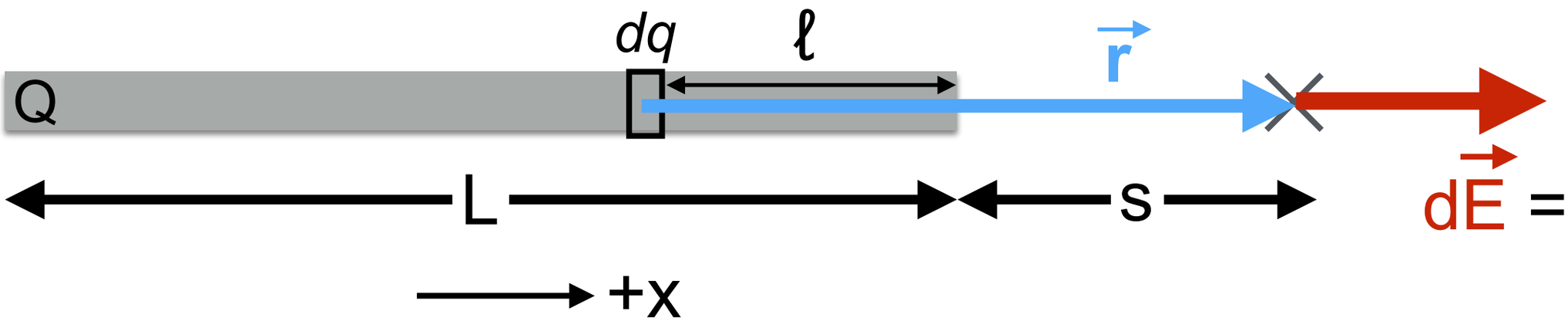
Charged Volume



$\rho_{uniform} = Q / V \quad dq = \rho \, dV$

Line of Charge - Redo!

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$d\vec{E} = K\frac{dq}{r^2}\hat{r}$$

What is the distance vector $\vec{r}$?

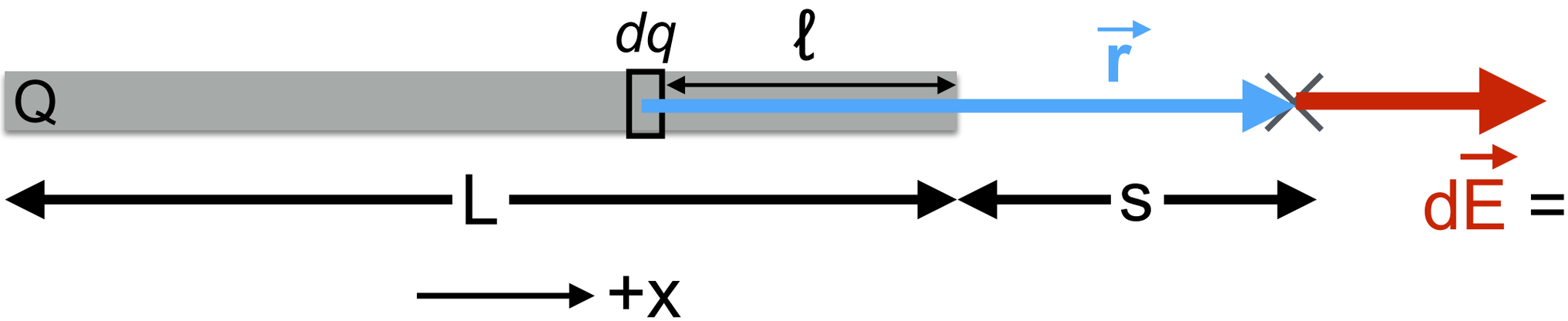
It points from dq to the point we are calculating the electric field.

1. $\vec{r} = \ell\hat{i}$
2. $\vec{r} = (\ell+s)\hat{i}$
3. $\vec{r} = (\ell-s)\hat{i}$
4. $\vec{r} = (\ell-s+L)\hat{i}$
5. $\vec{r} = (L-\ell+s)\hat{i}$

Line of Charge - Redo!

Distance Vector

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$d\vec{E} = K\frac{dq}{r^2}\hat{r}$$

$\vec{r} = (\ell+s)\hat{i}$

Magnitude : distance from dq to the point

$$|\vec{r}| = (\ell+s) = r$$

Direction : same direction as dE (turn into unit vector)

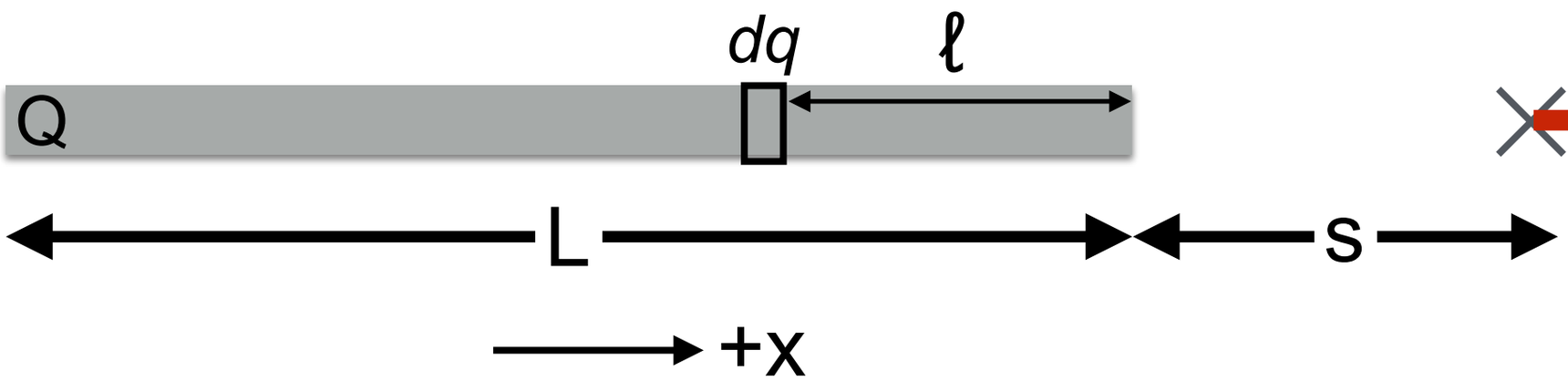
$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{(\ell+s)\hat{i}}{(\ell+s)} = \hat{i}$$

$$d\vec{E} = K\frac{dq}{(\ell+s)^2}\hat{i}$$

Line of Charge - Redo!

Charge Distribution

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$d\vec{E} = K\frac{dq}{(\ell+s)^2}\,\hat{i}$$

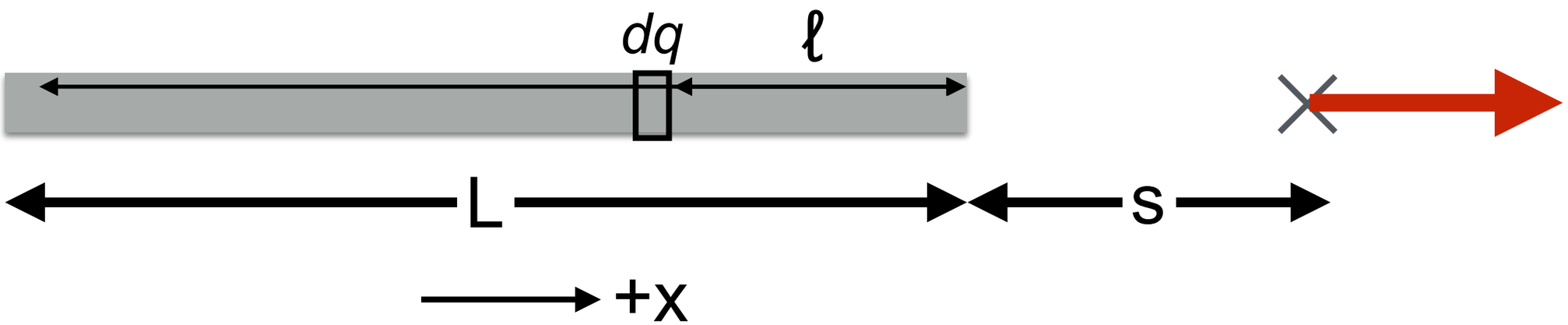
What charge distribution should we use?

1. Line of Charge, so that $dq = \lambda\, d\ell$
2. Area of Charge, so that $dq = \eta\, dA$
3. Volume of Charge, so that $dq = \rho\, dVol$
0. Don't wait on my response

Line of Charge - Redo!

Integrating Electric Field

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



From one section of the line : $d\vec{E} = k \dfrac{\lambda \, d\ell}{(\ell+s)^2} \hat{i}$

From whole line : $\vec{E} = \int d\vec{E}$

$$\vec{E} = \int k \frac{\lambda \, d\ell}{(\ell+s)^2} \hat{i} = k \lambda \hat{i} \int_0^L \frac{d\ell}{(\ell+s)^2}$$

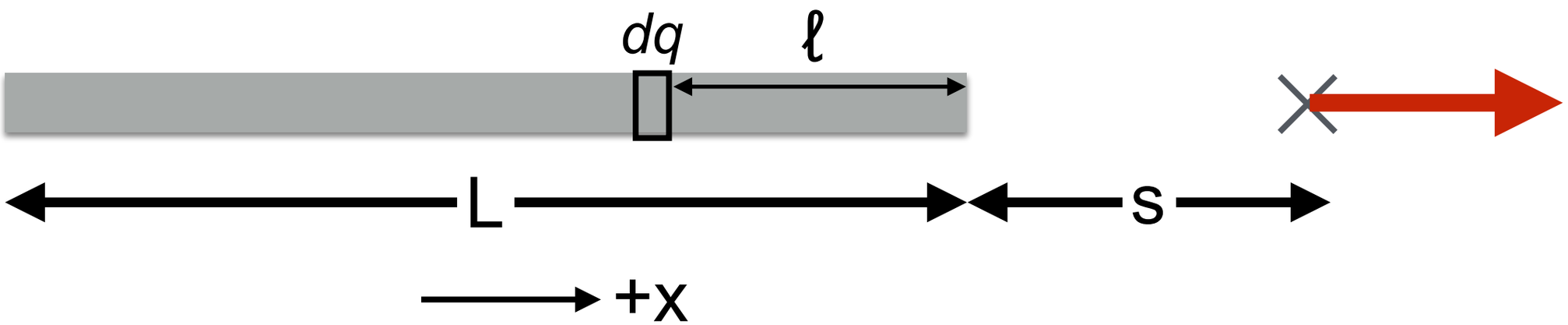
Closest distance : $\ell = 0$

Farthest distance : $\ell = L$

Line of Charge - Redo!

Integration by Substitution

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$\vec{E} = k\lambda\hat{i}\int_0^L \frac{d\ell}{(\ell+s)^2} \longrightarrow k\lambda\hat{i}\int_s^{L+s} \frac{dx}{x^2}$$

Substitution: $x = \ell + s$

$x = L+s$

$x = 0+s$

$$\frac{dx}{d\ell} = 1 \longrightarrow dx = d\ell$$

Line of Charge - Redo!

Integrate!

What is $\int \frac{dx}{x^2}$

$$\int x^{-2}\,dx = \frac{1}{-2+1}\,x^{-2+1}$$

$$= \frac{1}{-1}\,x^{-1} = \frac{-1}{x}$$

1. $1 / x$
2. $-1 / x$
3. $1 / x^3$
4. $-2 / x^3$
5. $-3 / x^3$

$$\int_s^{L+s} x^{-2}\,dx = \left.\frac{-1}{x}\right|_s^{L+s}$$

$$= \frac{-1}{L+s} - \frac{-1}{s}$$

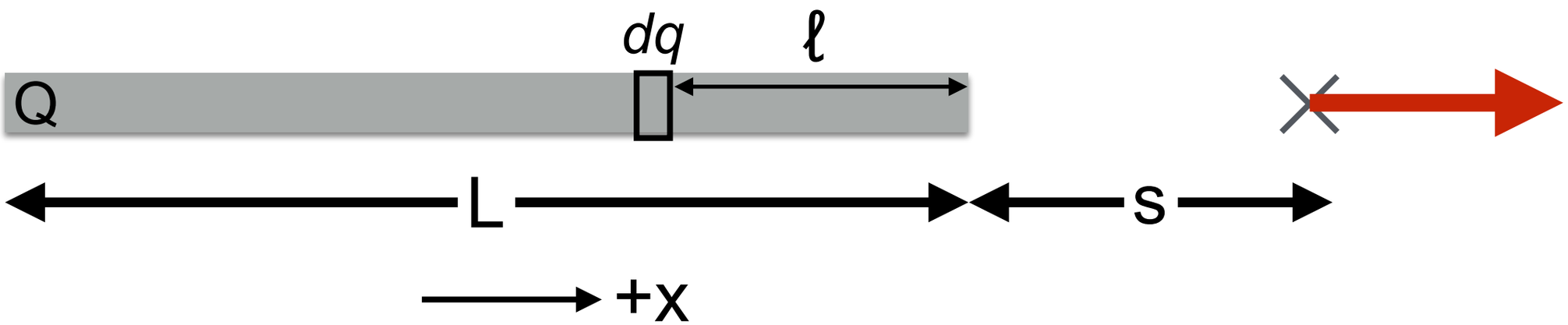
$$= \frac{-s}{s(L+s)} + \frac{L+s}{s(L+s)}$$

$$= \frac{-s + L+s}{s(L+s)} = \frac{L}{s(L+s)}$$

Line of Charge - Redo!

Finishing the Problem!

A line with length L and total charge Q creates an electric field. **Derive** the vector equation for the field a distance s from the end of the line in the x-direction.



$$\vec{E} = k\,\lambda\,\hat{i}\int_0^L \frac{d\ell}{(\ell+s)^2} = k\,\lambda\,\hat{i}\,\frac{L}{s(L+s)} = k\,\frac{Q}{\cancel{L}}\,\hat{i}\,\frac{\cancel{L}}{s(L+s)}$$

Q>0, $\vec{E}$ is toward the right

Q<0, $\vec{E}$ is toward the left

$$\vec{E} = k\,\frac{Q}{s(L+s)}\,\hat{i}$$

Steps for Continuous Fields

Draw a picture and setup coordinates

Chose a tiny charge dq on the object (no special locations)

Figure out the <u>distance vector</u> $\vec{r}$ from dq to your point

Use the distance vector to get distance and direction

Write down dE for that point : $d\vec{E} = K \frac{dq}{r^2} \hat{r}$

Change dq to a distribution and differentials

Figure out the limits

Integrate and simplify?